Explainability in Reinforcement Learning

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Chapter 1

Latent Space

After presenting, in the previous chapter, the fundamental principles of deep learning Chapter **??**, we now focus on one of the most important outcomes of this learning process: latent representations. When a deep neural network processes an input, it does not directly produce a final prediction. Each hidden layer also generates an intermediate representation of the information, as illustrated in Figure **??**. These internal representations progressively transform the input data in a way that is adapted to the task being solved. The purpose of this chapter is to study these intermediate representations in order to understand the type of information they contain and what can be extracted from them. The study of latent representations is a relatively recent research field. Although rapidly expanding, this domain still lacks a fully unified terminology. Different expressions coexist depending on the scientific community, such as *latent semantic analysis*, *representation learning*, *embedding analysis*, or *concept discovery.* In this manuscript, we will use the general term **latent representation analysis** to encompass these approaches.

- We then discuss the notions of concept, representation, and semantics, which form the theoretical foundations of latent space analysis.
- First, we introduce the main terminology and clarify the meaning of the concepts used throughout this manuscript.

Once these foundations are established, we present the principal approaches used to identify and extract the concepts encoded within neural networks. To the best of our current knowledge, four major families of methods can be distinguished:

- the use of **probes** to detect the presence of a given concept within a latent representation;
- the identification of **semantic directions** in latent spaces, corresponding to transformations interpretable by humans;
- the study of the **superposition hypothesis**, as well as the use of **Sparse Autoencoders** to isolate individual concepts;
- **deep clustering** approaches, which aim to identify groups of representations sharing a common semantic meaning.

1.1 Clarification of Notions: Feature, Representation, Concept, and Semantics

In the deep learning literature, the notions of *feature*, *representation*, *concept*, and *semantics* are often used with varying degrees of precision across communities (machine learning, computer vision, and interpretability). This section provides a conceptual clarification grounded in foundational works in representation learning.

1.1.1 Feature

A **feature** is a numerical variable derived from observed data, constituting the elementary unit of information processed by a learning model. In classical machine learning, features were manually engineered. In deep learning, they are now automatically learned at multiple levels of abstraction within the network [**?**]. Formally, a feature can be interpreted as a component of an input vector $x \in \mathbb{R}^d$ or of an intermediate activation $h^{(l)} \in \mathbb{R}^{d_l}$. Thus, features correspond to local, scalar-valued quantities forming the basis of learned representations.

1.1.2 Representation

A **representation** refers to a transformed encoding of data in a latent space, obtained through a composition of parameterized functions:

$$h^{(l)} = f^{(l)}(h^{(l-1)})$$

Each layer of a deep neural network learns such a transformation, producing a hierarchy of representations of increasing abstraction. According to Bengio et al. [**?**], deep learning aims at learning distributed representations that capture explanatory factors of variation in the data.

These representations are typically:

- distributed (each dimension contributes to multiple factors),
- hierarchical,
- task-dependent.

From a geometric perspective, a representation corresponds to a learned embedding of the data into a latent space where relevant factors become more explicitly organized.

1.1.3 Concept

A **concept** is defined as a semantic abstraction corresponding to a stable regularity in the real world, independent of any particular encoding within a model. In neural networks, concepts are not explicitly represented but rather emerge from the structure of learned representations. Work in interpretability has shown that concepts can often be associated with directions or subspaces in latent representations [**?**, **?**]. For instance, a concept such as *"dog"* may correspond to multiple internal configurations without a unique or bijective mapping to a specific neuron or feature. Thus, concepts are best understood as emergent semantic entities inferred from learned representations rather than explicitly encoded variables.

1.1.4 Semantics

Semantics refers to the relationship between an internal representation and its interpretation in terms of meaning in the real world. In deep learning, semantics are typically implicit and must be inferred a posteriori. They correspond to the alignment between geometric structures in representation spaces and human-interpretable concepts. Representation learning has shown that certain geometric properties (e.g., clustering structure, linear separability, vector directions) correlate with semantic regularities [**?**, Goodfellow et al., 2016].

1.1.5 Summary

1.2 Fundamental Definitions

1.2.1 Latent Activation

A **latent activation** (also referred to as a *hidden activation* or *intermediate activation*) denotes the output of a hidden layer $l \in \{2, \ldots, L-1\}$ of a neural network parameterized by $\hat{f}_\theta$, given an input x. This activation is represented by a vector:

$$h_l = [a_1, a_2, \ldots, a_{n_l}] \in \mathbb{R}^{n_l},$$

where n_l denotes the number of neurons in layer l. We denote by ϕ_l the function mapping an input x to its latent activation at layer l: $\phi_l : \mathcal{X} \to \mathbb{R}^{n_l}$ The function ϕ_l implicitly contains all transformations applied from the input layer up to layer l. It can therefore be expressed recursively as the composition of all preceding layers. At initialization, since the network parameters are randomly initialized, latent activations do not carry any particular meaning. They merely correspond to random projections of the input data.

1.2.2 Latent Representation

During training, the network parameters are progressively adjusted so that latent activations become informative with respect to the target task. Consequently, for an input x, the vector $\phi_l(x)$ evolves in such a way that it captures the relevant characteristics required for the network to produce the expected output. Once training is completed, these intermediate activations are referred to as **latent representations** (or *embeddings*). The term "representation" emphasizes that these vectors are no longer simple random projections, but instead encode structured and task-relevant information. In other words, they constitute a new description of the data, optimized to allow the network $\hat{f}_\theta$ to correctly predict the output y associated with the input x.

1.2.3 Latent Space

A latent representation only acquires meaning when considered relative to other representations. Let us consider a dataset:

$$\{x_1, x_2, \ldots, x_m\}.$$

After training, projecting these samples through the function ϕ_l produces the set:

$$\{\phi_l(x_1), \phi_l(x_2), \ldots, \phi_l(x_m)\}.$$

This structured set of vectors is called the **latent space** (also referred to as the *embedding space* or *latent feature space*). The relative organization of representations within this space reflects how the network distinguishes different inputs and encodes the information necessary for prediction. It is important to note that the term “latent space” should not be understood solely in the mathematical sense of a subspace of $\mathbb{R}^{n_l}$. More broadly, it refers to the semantic structure induced by the representations and the relationships between them. This structure forms the foundation of the interpretation methods studied throughout this chapter.

Bibliography

[Goodfellow et al., 2016] Goodfellow, I., Bengio, Y., Courville, A., and Bengio, Y. (2016). Deep learning, volume 1. MIT press Cambridge.